Methodological Revision on Causal Depth and Spatial Indexing

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> Version Note: This version supersedes v1.5.1 (Causal Topology Edition). The earlier heuristic sketch treating Planck-scale quantities as native pixel or atomic clock cycles of the underlying network has been replaced. Reconstructed fully under SRE Axioms v1.6 and the dissipation-compensation duality framework, Planck quantities are defined as **emergent ontological ultraviolet boundaries**, rather than intrinsic granular parameters of causal nodes.

I. The Non-Axis Nature of Time: Causal Logical Depth as Dissipation-Compensation Book-Keeping

Time is neither a fundamental dimension nor a pre-given physical axis. Within the SRE framework, time is not a hardware clock native to the underlying network. Instead, it is the emergent manifestation of causal logical depth in the process of informational state change, after the stacking of reciprocal-measurement events.

- **Causal Triggers and Book-Keeping Flux**: The sense of temporal "flow" is the macro-level mapping of sequential book-keeping for a large ensemble of reciprocal-measurement events. Planck time τ_P is an emergent ontological ultraviolet boundary, setting the lower cost bound for logical operations eligible for instantiation. **It is not an intrinsic atomic cycle clock of the causal network**. Causal differences falling below this cost threshold remain in the pending-for-instantiation state within causal space and cannot generate new temporal increments.
- **Topological Overhead for Global Consistency**: Causal links themselves are discrete logical associations, yet the perception of duration arises from the topological routing overhead required for both parties in reciprocal measurement to finish dissipation evaluation and compensation settlement. Time may therefore be understood as the topological metabolic cost incurred by the causal network to maintain global self-consistency within the rendering layer. This book-keeping process is continuously driven by reciprocal-measurement mediated via Möbius light-residual manifolds.

II. Spatial Indexing: Rendered Projection of Causal Connectivity

Three-dimensional space is no pre-existing physical entity. It is a dimensionality-reduced rendering-projection system generated through dissipation-compensation duality for the causal network to classify and navigate discrete causal relationships.

- **Causal Nodes and Instantiation Boundaries**: Physical distance is no longer simply equivalent to the count of intermediate logical steps required for node synchronization. It is a topological book-keeping quantity jointly evaluated from the topological dissipation tensor and the compensation operator. The Planck length l_P is an emergent ontological ultraviolet boundary that marks the instantiable causal-connectivity boundary of the physical rendering layer, **rather than the minimal pixel granularity of the underlying causal network**. Causal differences below this boundary will not be projected into distance-related spatiotemporal states of affairs. Interpreted heuristically from a physical-image perspective: distance reflects the degree of coherence degradation between two topological residual manifolds. Higher coherence corresponds to lower mutual informational dissipation, which in turn yields a shorter macroscopic spatial distance.

- **Dimensional Projection**: Our perception of three-dimensional geometry arises from data reduction. The system projects complex high-dimensional causal topologies onto effective spatial ranks unlocked across the BBP spectral phase transition, yielding simplified spatial metrics for reciprocal-measurement interactions among causal nodes.

III. Spatiotemporal Interchangeability: Causal Re-mapping

Distinguishing time as **dissipation-book-keeping depth** and space as **rendered projection of causal connectivity**, SRE opens a theoretical path toward spatiotemporal interchangeability.

- Redefinition of Displacement: Physical motion shall no longer be interpreted as continuous translation inside a pre-supposed background container. Instead, it consists of dynamic re-weighting of associative links among discrete causal nodes, driven by adjustments of compensation flows triggered by local variations of informational dissipation.
- **Apparent Instantaneous Leap (Causal Re-mapping)**: The theoretical objective is to perform “causal re-mapping”: bypass sequential intermediate causal steps and establish direct topological associative links between two sets of quantum-evidential nodes. Subject to network-stability and instantiation-boundary constraints, this operation minimizes traversed causal book-keeping depth and thereby produces an apparent instantaneous-leap effect within the physical rendering layer. This process cannot violate the self-consistency constraints of the underlying causal network; it remains purely an apparent effect at the rendering layer and does not break causal self-consistency.

Key Rewrite Reference Memorandum

1. Removed content inherited from old v1.5.1: *Planck time as the atomic cycle for single causal triggers, Planck length as indexing pixel granularity, distance as count of intermediate logical steps*.
2. Retained and reconstructed core conceptual kernels:
   - Time = emergence of causal logical depth, where logical depth originates from **dissipation-compensation book-keeping of reciprocal measurements**, not a native hardware clock.
   - 3D space = dimension-reduced rendering projection of high-dimensional causal topology.
   - Displacement = re-weighting of causal associative links.
   - Causal re-mapping: establishing direct links to bypass intermediate steps, producing apparent instantaneous-leap effects at rendering layer, with added constraint conditions.
3. Uniform terminology across the whole manuscript: **emergent ontological ultraviolet boundary, dissipation-compensation duality, reciprocal measurement, Möbius light-residual manifold, BBP spectral phase transition, pending-for-instantiation state, physical rendering layer**.
